

Bhavyasri Sanisetty

Machine Learning Engineer | Python | Gen-AI Systems | FastAPI

GitHub - Sanisetty-bhavya • LinkedIn - bhavyasri-sanisetty-31948424b • Portfolio

• bhavyasrisanisetty@gmail.com • +91-8519862296

- Published peer-reviewed ML research (Scientific Reports — Nature Portfolio, 2025) improving forecasting error by 15–20%
- Developed and deployed ML inference backend serving real-time predictions via REST APIs
- Built JWT-based multi-factor authentication system securing protected endpoints

Flagship Production System

Smart Agriculture ML Platform — Deployed Backend Service

- Designed and deployed a FastAPI-based ML backend providing crop recommendation, weed detection, and pest classification using environmental and image inputs.
- Evaluated 20 ML/DL models on a **2,200-record dataset (22 classes)** and deployed a Voting Classifier ensemble improving prediction accuracy by 15–20%.
- Built large-scale image classification pipelines covering **232 species using 65k+ labeled images (ResNet50, CNN, ViT)**
- Developed modular FastAPI inference APIs with schema validation and structured JSON responses for reusable ML services.
- Secured APIs using JWT authentication and deployed backend on Railway; validated live inference workflows via HTTP testing.

Published Research —Scientific Reports (Nature Portfolio), 2025

- Led implementation of a hybrid **ANN + Transformer** forecasting framework integrating fuzzy preprocessing to model noisy multivariate energy load data and long-range temporal dependencies.
- Formulated data preprocessing and feature engineering pipelines to process real-world hourly energy consumption variables including electricity, gas, HVAC, and auxiliary signals.
- Benchmarked the hybrid model against traditional and standalone baselines (Random Forests, Decision Trees, Linear Regression) achieving **~15–20% reductions in RMSE and MAE** and strong generalization.
- Developed reusable training and evaluation workflows in **Python and PyTorch** enabling scalable experimentation across model variants.
- Validated model effectiveness on **real energy datasets** supporting improved load forecasting insights for grid planning.

Financial Document Intelligence Assistant — RAG Gen-AI System

- Built **Retrieval-Augmented Generation (RAG)** pipeline using FastAPI, FAISS, and transformer **embeddings** to query financial documents.
- Implemented document ingestion, **semantic chunking**, and **vector indexing** for multi-document search.
- Generated grounded responses with **citation retrieval** to reduce **LLM hallucinations**.
- Developed **REST APIs** for document upload and natural-language querying.
- Evaluated retrieval quality using **similarity scoring** and **top-k retrieval**.

Additional Engineering Project

Multi-Factor Authentication Backend Service

FastAPI · JWT · OTP · REST APIs

- Engineered JWT + OTP authentication securing protected API endpoints with middleware-enforced access control.
- Implemented multi-step authentication flow (login → OTP verification → active session).
- Enforced token expiry validation and authorization checks on protected endpoints.
- Implemented role-protected routes with token validation and session state management.

Professional Experience

Salesforce Platform Configuration

May 2024 – Jun 2024

- Worked within a structured CRM environment configuring validation rules and workflow automation simulating enterprise data operations.

Technical Skills

Programming: Python

Gen-AI & NLP: Retrieval Augmented Generation (RAG), Embeddings, Semantic Search, Prompt Engineering

Machine Learning: PyTorch, Scikit-learn, Model Evaluation, Feature Engineering, Time-Series Modeling

Backend: FastAPI, REST APIs, JWT Authentication, Request Validation

Databases: MongoDB, SQLite, FAISS (Vector Database)

Tools & Deployment: Git, Linux, Postman, Railway

Education

B.Tech Computer Science & Engineering — SRM University (2025)